

Notes: Campbell, Giglio & Pathak (AER, 2011)

Forced Sales and House Prices

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Transaction data and sample	4
3.2	Forced-sale categories	4
3.3	Timing windows	4
3.4	Geography	5
3.5	Main outcome variables	5
4	Models and empirical specifications	5
4.1	Static hedonic regression	5
4.2	Disaggregated forced-sale indicators	5
4.3	Value-component interaction model	6
4.4	ZIP code VAR for neighborhood price dynamics	6
4.5	Local foreclosure spillover specification	6
5	Identification and empirical design	6
5.1	Main threat: forced sales are not random	6
5.2	Identification of forced-sale discounts	7
5.3	Why death-related sales are useful	7
5.4	Why bankruptcy-related sales are different	7
5.5	Why foreclosure discounts are hard to interpret	7
5.6	Identification of neighborhood spillovers	8
5.7	What remains non-experimental	8
6	Tables and figures	8
6.1	Table 1: Frequency and Timing of Forced Sales; Figure 1: Geographic Distribution of Percentage of Housing Transactions That Are Forced Sales by ZIP Code	8
6.2	Table 2: Price Discount for Forced Sales; Figure 2: Forced-Sales Discount and Time between Sale and Event	9
6.3	Table 3: Price Discounts and Value Components	10
6.4	Table 4: VAR for Neighborhood House Prices	10
6.5	Table 5: Spillover Estimates of Foreclosures	11
7	Short summary	12

8	Conclusion	13
8.1	Core conclusion	13
8.2	Economic intuition	13
8.3	Interpretation of the discount estimates	13
8.4	Identification takeaway	13
8.5	Policy implication	13
8.6	Limitations	14
8.7	Takeaway	14

1 Big picture

- **Question.** How large are price discounts for urgent or forced house sales, and do foreclosures depress the prices of nearby houses?
- **Setting.** Massachusetts residential transactions from 1987 through the first quarter of 2009, a period that includes an early-1990s downturn and the 2007–2009 foreclosure wave.
- **Core idea.** Housing is illiquid: houses are heterogeneous, indivisible, expensive, and costly to maintain when vacant. Urgent sellers may accept lower prices than ordinary sellers.
- **Forced-sale categories.** The paper identifies sales linked to seller death, personal bankruptcy, and foreclosure. Death and bankruptcy are matched by names and addresses; foreclosures are identified through deed and auction records.
- **Main forced-sale discounts.** Forced sales trade at large discounts relative to otherwise similar houses in the same census tract-year. The biggest discount is for foreclosures, roughly 27–31 percent in log-price terms depending on specification. Death-related sales have smaller discounts, about 5–7 percent, and bankruptcies are smaller still.
- **Mechanisms.** Death-related discounts look partly like poor maintenance, especially for older sellers and houses where the structure accounts for a larger share of value. Bankruptcy discounts look more like a liquidity effect. Foreclosure discounts combine urgency, vacancy, vandalism, and lender liquidation incentives.
- **Neighborhood dynamics.** At the ZIP code level, forced-sale prices mean-revert strongly, but unforced-sale prices are much closer to a random walk. This weakens the view that forced sales are simply revealing future market prices.
- **Foreclosure spillovers.** At very local distances, foreclosures appear to lower prices of nearby houses. The estimated effects are strongest for foreclosures within 0.1 mile and for foreclosures occurring before the sale rather than after it.
- **Economic takeaway.** Forced sales reveal the illiquidity of residential real estate. The individual forced-sale discount is large, and nearby foreclosures can create local externalities even when ZIP-level price dynamics look relatively efficient.

2 Incremental contribution of the paper

- **From anecdote to measurement.** The paper converts the broad idea that foreclosures sell cheaply into a transaction-level measurement exercise using more than 1.8 million house sales.
- **Separating different forms of urgency.** Instead of treating all distressed transactions as one category, the paper separates death-related sales, bankruptcy-related sales, and foreclosure-related sales. This matters because the underlying mechanisms differ across categories.
- **Administrative event matching.** The paper links housing transactions to independent data on deaths, bankruptcy filings, and foreclosure deeds/auctions. This is an important step beyond using only transaction prices or broad neighborhood foreclosure rates.
- **Within-neighborhood hedonic identification.** The main discount estimates use census tract-year fixed effects and housing controls, which compare forced and unforced sales in very similar local markets at the same time.
- **Mechanism tests through heterogeneity.** The paper does not stop at estimating a discount. It studies timing relative to death/bankruptcy, number of sellers, housing type, and building/land/neighborhood value components to distinguish maintenance, liquidity, and vandalism channels.

- **Dynamic distinction between bid-price and market-price views.** The ZIP-level VAR asks whether forced sales forecast unforced prices. The result that unforced prices are hard to predict suggests forced sales often reveal a temporary bid-side discount rather than the efficient future price.
- **Localized spillover design.** The paper introduces a highly local foreclosure spillover analysis based on distance and timing. The before-versus-after and close-versus-far comparisons are designed to separate true spillovers from common neighborhood shocks.
- **Policy relevance.** The results speak directly to foreclosure externalities and mortgage market policy. If nearby foreclosures depress prices, then foreclosure waves may generate feedback loops through negative equity and refinancing constraints.

Incremental contribution in one sentence. The paper shows that forced-sale discounts are large and heterogeneous, and that foreclosure spillovers are highly localized, thereby moving the literature from aggregate foreclosure-price correlations to micro-level evidence on illiquidity, maintenance, lender urgency, and neighborhood externalities.

3 Data and measurement

3.1 Transaction data and sample

The core data come from Massachusetts residential housing transactions from 1987 through March 2009. The transaction records include sale prices, dates, property characteristics, address information, housing type, and tract/ZIP geography. The final sample contains more than 1.8 million transactions.

Interpretation: The data are unusually useful because Massachusetts experienced both an early-1990s foreclosure episode and the later foreclosure wave. This gives repeated variation in forced-sale incidence over time and across neighborhoods.

3.2 Forced-sale categories

A transaction is classified as forced if it is linked to one of three events:

- **Death:** the seller is matched to the Social Security Death Master File within three years before or after the sale.
- **Bankruptcy:** the seller is matched to personal bankruptcy filings within three years before or after the sale.
- **Foreclosure:** the property transfer is foreclosure-related; for unsuccessful auctions, the subsequent sale by the mortgage lender is treated as a forced sale.

Interpretation: The three categories are not equivalent. Death-related sales may reflect maintenance and life-cycle effects. Bankruptcy is closer to financial distress. Foreclosure involves lender ownership, vacancy, possible vandalism, and strong incentives for rapid liquidation.

3.3 Timing windows

For death and bankruptcy, sales are classified as forced if they occur within three years before or after the event. For foreclosures, the relevant forced-sale window is tied to the foreclosure auction closure and subsequent lender sale.

Interpretation: The broad timing window allows the paper to capture transactions that occur in anticipation of, or in response to, the forcing event. Later analysis uses timing variation to ask whether the discount looks like urgency or like unobserved house quality.

3.4 Geography

The paper uses several geographic units:

- census tract-year fixed effects in hedonic regressions;
- ZIP code-year panels for price dynamics;
- precise distance measures within 0.25 mile and 0.1 mile for foreclosure spillovers.

Interpretation: Each geography answers a different question. Tract-year fixed effects absorb local market conditions. ZIP code panels study neighborhood price dynamics. Very local distances test whether foreclosures create localized price externalities.

3.5 Main outcome variables

The main transaction-level outcome is log house price:

$$y_{ist} = \log(P_{ist}),$$

where i indexes a house transaction, s indexes census tract, and t indexes year.

Other outcomes include average forced and unforced prices at the ZIP-year level, the share of forced sales, and local counts of foreclosures near a transaction.

Interpretation: Log price coefficients can be read approximately as percentage discounts. For large negative coefficients such as foreclosure discounts, the exact percentage effect is $\exp(\lambda) - 1$.

4 Models and empirical specifications

4.1 Static hedonic regression

The main transaction-level hedonic specification is:

$$y_{ist} = \alpha_{st} + \beta' X_i + \lambda' F_i + \varepsilon_{ist}. \quad (1)$$

Here α_{st} are census tract-year fixed effects, X_i contains house characteristics, and F_i is a vector of forced-sale indicators.

Interpretation: The coefficient λ compares forced and unforced sales of similar houses in the same census tract and year. This is the main estimate of the forced-sale discount.

4.2 Disaggregated forced-sale indicators

The paper estimates separate coefficients for:

$$F_i \in \{\text{young death, old death, bankruptcy, foreclosure}\}.$$

It also estimates separate regressions by housing type:

$$\text{single-family, multifamily, condominium.}$$

Interpretation: Disaggregation matters because a foreclosure is not just a forced sale; it is also a vacant-house and lender-liquidation event. Death and bankruptcy are different types of urgency.

4.3 Value-component interaction model

To interpret mechanisms, the paper decomposes predicted house value into building, lot size, and tract-year components, then interacts forced-sale indicators with standardized components:

$$y_i = \alpha_{st} + \beta' X_i + \lambda' F_i + \gamma'_B(F_i \times B_i) + \gamma'_L(F_i \times L_i) + \gamma'_N(F_i \times N_i) + u_i. \quad (2)$$

Interpretation: If discounts are driven by poor maintenance of structures, they should be larger when the building component is more valuable. If discounts reflect fixed protection or vandalism costs, they may be proportionally larger for low-value properties.

4.4 ZIP code VAR for neighborhood price dynamics

At the ZIP-year level, the paper studies the vector:

$$Z_{ft} = \left(s_{ft}, \Delta p_{ft}^f, p_{ft}^u - p_{ft}^f, \Delta p_{ft}^u \right),$$

where s_{ft} is the share of forced sales, p^f is average forced-sale log price, and p^u is average unforced-sale log price.

A stylized VAR is:

$$Z_{ft} = AZ_{f,t-1} + \mu_f + \tau_t + e_{ft}. \quad (3)$$

Interpretation: This model asks whether forced-sale discounts predict future unforced-sale price declines or instead mean-revert mainly through forced-sale prices.

4.5 Local foreclosure spillover specification

The spillover regression adds distance-weighted counts of nearby foreclosures before and after the transaction:

$$\log P_i = \alpha_{st} + \beta' X_i + \lambda' F_i + g_B(N_{i,close}^B, N_{i,far}^B) + g_A(N_{i,close}^A, N_{i,far}^A) + \varepsilon_i. \quad (4)$$

Here B denotes foreclosures before the sale, A denotes foreclosures after the sale, close refers to the 0.1-mile neighborhood, and far refers to the remaining area inside 0.25 mile.

Interpretation: The causal logic comes from comparing past versus future foreclosures and very close versus less close foreclosures. A stronger effect from past and close foreclosures is more consistent with a spillover than with broad unobserved neighborhood trends.

5 Identification and empirical design

5.1 Main threat: forced sales are not random

Forced sales may involve houses, owners, or neighborhoods that differ systematically from ordinary sales. Death-related sales may involve older owners with poorly maintained homes. Bankruptcy and foreclosure may be linked to economic distress. Foreclosures may occur in neighborhoods already declining.

Interpretation: The paper therefore treats the raw price difference between forced and unforced sales as descriptive, and relies on fixed effects, controls, timing variation, and distance-based tests to sharpen interpretation.

5.2 Identification of forced-sale discounts

The static hedonic design identifies forced-sale discounts from comparisons of forced and unforced transactions within the same census tract-year:

$$\lambda = E[y|F = 1, X, \alpha_{st}] - E[y|F = 0, X, \alpha_{st}].$$

The key controls are:

- detailed house characteristics;
- census tract-year fixed effects;
- separate indicators for different forced-sale triggers;
- standard errors clustered at the census tract-year level.

Interpretation: Tract-year fixed effects are powerful because they absorb local time-varying housing-market conditions. The remaining identification comes from forced and unforced sales within a narrowly defined local market at the same time.

5.3 Why death-related sales are useful

Death-related transactions are not fully exogenous, but they are useful because the timing and seller age provide mechanism tests:

- older-death discounts are larger than young-death discounts;
- timing around death is relatively insensitive;
- discounts are larger when the building component is more important.

Interpretation: These facts push the interpretation away from pure sale urgency and toward poor maintenance by older homeowners.

5.4 Why bankruptcy-related sales are different

Bankruptcy-related discounts are smaller than foreclosure discounts and show a timing pattern closer to a liquidity event: the largest discounts appear around the period after bankruptcy filing.

Interpretation: This suggests that bankruptcy discounts are more related to financial pressure and urgency, not simply to long-standing physical deterioration.

5.5 Why foreclosure discounts are hard to interpret

Foreclosures are the most important but also the most endogenous category. A foreclosed property may sell cheaply because:

- the lender wants to liquidate quickly;
- the house is vacant and costly to protect;

- the house may have been vandalized;
- the neighborhood may be declining;
- the foreclosure itself may depress the neighborhood.

Interpretation: The paper cannot fully separate all these channels in the static hedonic regression. It therefore uses value-component heterogeneity and local spillover tests as additional evidence.

5.6 Identification of neighborhood spillovers

The spillover analysis compares nearby foreclosures before the sale to nearby foreclosures after the sale. Future foreclosures proxy for local conditions that are correlated with foreclosure risk but cannot themselves have already affected the transaction.

The identifying contrast is:

$$\delta^{spillover} \approx \delta_{before} - \delta_{after}.$$

The second contrast is distance:

$$\delta_{close} \quad \text{versus} \quad \delta_{far}.$$

Interpretation: If unobserved neighborhood decline drives both house prices and foreclosures, it should affect nearby and slightly less nearby foreclosures similarly, and it should show up in future foreclosures as well. The finding that past very-close foreclosures matter more strengthens the spillover interpretation.

5.7 What remains non-experimental

The authors explicitly note that the results cannot be definitive on causality from foreclosures to house prices. There is no clean instrument that shifts foreclosures while otherwise leaving house prices unaffected.

Interpretation: The evidence is best read as a careful quasi-causal design rather than a randomized experiment. The most persuasive parts are the timing, distance decay, and robustness evidence, not the cross-sectional foreclosure-price correlation alone.

6 Tables and figures

6.1 Table 1: Frequency and Timing of Forced Sales; Figure 1: Geographic Distribution of Percentage of Housing Transactions That Are Forced Sales by ZIP Code

Read:

- Table 1 shows 1,831,393 transactions, with 6.1 percent classified as forced.
- Foreclosures account for 3.5 percent of all transactions, deaths 1.8 percent, and bankruptcies 0.8 percent.
- Foreclosures spike in the crisis period: 14.0 percent of transactions in 2008 and 25.7 percent in 2009 Q1.
- Panel B shows that most foreclosure-related sales occur after the foreclosure auction, while death and bankruptcy sales are spread more evenly around the event.
- Figure 1 shows that forced sales are not geographically random. They cluster in certain ZIP codes, especially in darker-shaded areas.

Interpretation: The frequency table establishes the sample and the importance of foreclosure waves. The map explains why simple comparisons are misleading: forced sales occur disproportionately in lower-price or distressed neighborhoods. This motivates the tract-year fixed effects in the hedonic regressions.

Economic message: Forced sales are common enough to matter for market prices, but their uneven geography means that identification requires careful local controls.

TABLE 1—FREQUENCY AND TIMING OF FORCED SALES

Year	Total observations	Deaths	Bankruptcies	Foreclosures	Total forced
<i>Panel A. Number of forced transactions by year</i>					
1987	87,257	1.1%	—	0.0%	1.1%
1988	78,461	0.9%	—	0.0%	0.9%
1989	65,728	0.9%	—	0.3%	1.2%
1990	54,062	1.0%	—	1.1%	2.1%
1991	57,013	1.1%	0.1%	5.2%	6.4%
1992	68,471	1.2%	0.2%	8.2%	9.6%
1993	74,556	1.6%	0.3%	9.4%	11.4%
1994	81,058	1.8%	0.5%	8.3%	10.5%
1995	75,909	1.8%	0.6%	7.0%	9.3%
1996	84,046	1.6%	0.7%	4.9%	7.3%
1997	90,163	1.8%	0.8%	4.3%	6.9%
1998	99,770	1.9%	0.9%	3.0%	5.7%
1999	103,247	1.8%	1.1%	2.3%	5.2%
2000	95,036	1.9%	1.1%	1.8%	4.8%
2001	89,555	2.0%	1.2%	1.4%	4.5%
2002	92,582	2.2%	1.2%	1.2%	4.6%
2003	94,692	2.3%	1.4%	0.7%	4.5%
2004	105,630	2.5%	1.4%	0.7%	4.6%
2005	101,929	2.4%	1.3%	0.8%	4.5%
2006	86,243	2.3%	1.3%	1.6%	5.2%
2007	77,526	2.2%	0.9%	5.3%	8.4%
2008	60,483	1.9%	0.7%	14.0%	16.6%
2009(Q1)	7,976	2.1%	0.7%	25.7%	28.5%
Total	1,831,393	1.8%	0.8%	3.5%	6.1%
<i>Panel B. Timing of forced transactions relative to forcing event</i>					
Group	Death	Bankruptcy	Foreclosure		
Sale 3 years before event	12.9%	10.3%			
Sale 2 years before event	15.2%	10.1%			
Sale 1 year before event	20.6%	9.5%			
Sale 1 year after event	29.1%	30.8%	85.9%		
Sale 2 years after event	14.8%	22.1%	9.1%		
Sale 3 years after event	7.4%	17.2%	1.6%		

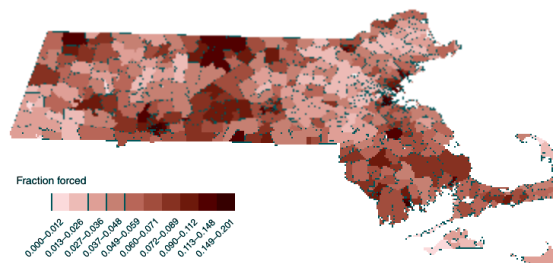


FIGURE 1. GEOGRAPHIC DISTRIBUTION OF PERCENTAGE OF HOUSING TRANSACTIONS THAT ARE FORCED SALES BY ZIP CODE

Figure 1: Geographic distribution of forced-sale shares by ZIP code. Darker areas have a higher fraction of forced transactions.

Table 1: Frequency and timing of forced sales. The table documents both the long-run forced-sale incidence and the sharp foreclosure surge in 2008–2009.

6.2 Table 2: Price Discount for Forced Sales; Figure 2: Forced-Sales Discount and Time between Sale and Event

Read:

- Panel A of Table 2 estimates an overall forced-sale discount of -0.197 log points for the full sample.
- Panel B separates triggers: young death (-0.053), old death (-0.069), bankruptcy (-0.035), and foreclosure (-0.314).
- The foreclosure discount is by far the largest, roughly corresponding to a 27 percent price discount using $\exp(-0.314) - 1$.
- Discounts differ by housing type: foreclosures are especially large for multifamily properties and condominiums.
- Panel C shows that death and bankruptcy discounts are larger when the property has only one seller, consistent with weaker bargaining or less backup decision-making.
- Figure 2 shows that death-related discounts are relatively persistent around the death event, while bankruptcy discounts are more concentrated around the event period.

Interpretation: Table 2 is the paper’s main forced-sale discount estimate. The key pattern is not just that forced sales are cheaper; it is that the size and structure of the discount vary strongly by trigger. Foreclosures look like severe liquidation events, death-related sales look partly like maintenance/owner characteristics, and bankruptcies look like financial distress with a smaller discount.

Economic message: The discount is not a single liquidity number. Residential illiquidity depends on who is selling, why they are selling, what type of house is sold, and whether the sale is a lender liquidation.

TABLE 2—PRICE DISCOUNT FOR FORCED SALES

	Full sample		Single family		Multifamily		Condominium	
	Estimate	Std err	Estimate	Std err	Estimate	Std err	Estimate	Std err
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A. All forced transactions</i>								
Forced (−3 years; +3 years)	−0.197	(0.002)	−0.159	(0.002)	−0.236	(0.004)	−0.224	(0.005)
<i>Panel B. Forced transactions by type</i>								
Death, young seller (−3; +3)	−0.053	(0.005)	−0.068	(0.006)	−0.027	(0.014)	−0.012	(0.012)
Death, old seller (−3; +3)	−0.069	(0.002)	−0.082	(0.003)	−0.062	(0.007)	−0.017	(0.006)
Bankruptcy (−3; +3)	−0.035	(0.003)	−0.042	(0.003)	−0.020	(0.008)	−0.033	(0.007)
Foreclosure	−0.314	(0.003)	−0.260	(0.003)	−0.344	(0.005)	−0.308	(0.006)
<i>Panel C. Forced transactions by number of sellers</i>								
<i>One seller</i>								
Death, young seller (−3; +3)	−0.083	(0.010)	−0.093	(0.012)	−0.057	(0.026)	0.007	(0.018)
Death, old seller (−3; +3)	−0.097	(0.004)	−0.107	(0.005)	−0.099	(0.013)	−0.025	(0.011)
Bankruptcy (−3; +3)	−0.064	(0.005)	−0.073	(0.006)	−0.024	(0.012)	−0.051	(0.010)
<i>Two sellers</i>								
Death, young seller (−3; +3)	−0.038	(0.006)	−0.056	(0.007)	−0.009	(0.016)	−0.022	(0.015)
Death, old seller (−3; +3)	−0.053	(0.003)	−0.070	(0.003)	−0.041	(0.008)	−0.013	(0.008)
Bankruptcy (−3; +3)	−0.017	(0.004)	−0.025	(0.004)	−0.016	(0.011)	−0.014	(0.010)
Number of observations	1,831,393		1,187,645		202,123		441,625	

Table 2: Price discount for forced sales. Coefficients are log-price discounts after house controls and census tract-year fixed effects.

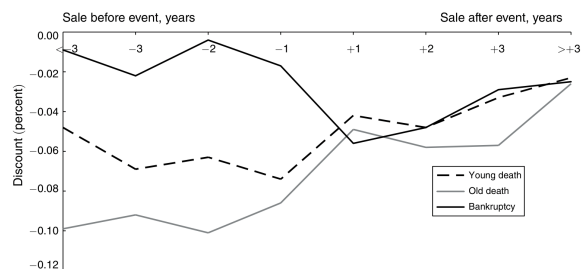


Figure 2: Forced-sale discount and timing relative to death or bankruptcy. The timing pattern helps separate urgency from persistent unobserved quality.

6.3 Table 3: Price Discounts and Value Components

Read:

- Table 3 interacts forced-sale indicators with the building component, lot-size component, and tract-year component of predicted house value.
- The main foreclosure effect remains large, about -0.283 log points in the full sample.
- Death-related discounts are larger when the building component is more valuable, especially for older sellers.
- Bankruptcy discounts vary less with building value but are more negative in high-price census tracts.
- Foreclosure discounts are larger for lower-value structures and lower-price areas, consistent with fixed protection/vandalism costs that are proportionally larger for cheaper properties.

Interpretation: This table is a mechanism table. The authors use value components to ask what kind of property is discounted more. If the death discount reflects poor maintenance, it should show up more strongly where the physical structure matters. If foreclosure discounts reflect fixed costs of vacancy or vandalism, they should be proportionally larger for cheaper homes.

Economic message: The forced-sale discount is partly an illiquidity discount and partly a real-asset-quality discount. Table 3 helps distinguish these channels.

6.4 Table 4: VAR for Neighborhood House Prices

Read:

- Table 4 aggregates prices to ZIP code-year cells and studies the dynamics of forced and unforced prices.
- Panel A shows reversal in forced-sale price growth: lagged forced-price growth predicts future forced-price growth with coefficient -0.072.
- A large forced-unforced price gap predicts forced-sale price increases, with coefficient 0.840 on the discount in the forced-price-growth equation.

TABLE 3—PRICE DISCOUNTS AND VALUE COMPONENTS

	Full sample		Single family		Multifamily		Condominium	
	Estimate	Std err	Estimate	Std err	Estimate	Std err	Estimate	Std err
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Main forced sale effect								
Death, young seller (−3; +3)	−0.055	(0.005)	−0.070	(0.007)	−0.023	(0.015)	−0.007	(0.012)
Death, old seller (−3; +3)	−0.071	(0.002)	−0.090	(0.003)	−0.062	(0.007)	−0.016	(0.007)
Bankruptcy (−3; +3)	−0.034	(0.003)	−0.040	(0.004)	−0.019	(0.010)	−0.033	(0.007)
Foreclosure	−0.283	(0.003)	−0.250	(0.003)	−0.281	(0.005)	−0.299	(0.007)
Forced sale effects interacted with building component								
Death, young seller (−3; +3)	−0.017	(0.007)	−0.006	(0.008)	0.032	(0.017)	0.023	(0.012)
Death, old seller (−3; +3)	−0.031	(0.004)	−0.014	(0.004)	0.004	(0.008)	−0.013	(0.008)
Bankruptcy (−3; +3)	−0.007	(0.004)	−0.003	(0.004)	0.007	(0.009)	−0.001	(0.008)
Foreclosure	0.041	(0.003)	0.039	(0.003)	−0.007	(0.005)	0.035	(0.006)
Forced sale effects interacted with lotsize component								
Death, young seller (−3; +3)	0.000	(0.006)	−0.005	(0.006)	0.015	(0.016)	−0.027	(0.015)
Death, old seller (−3; +3)	0.000	(0.003)	−0.019	(0.003)	0.016	(0.007)	0.009	(0.011)
Bankruptcy (−3; +3)	−0.006	(0.004)	−0.011	(0.004)	−0.014	(0.010)	0.003	(0.009)
Foreclosure	−0.008	(0.003)	0.004	(0.003)	0.017	(0.005)	−0.014	(0.007)
Forced sale effects interacted with tract-year component								
Death, young seller (−3; +3)	−0.011	(0.006)	−0.005	(0.006)	−0.013	(0.018)	−0.026	(0.013)
Death, old seller (−3; +3)	−0.004	(0.002)	0.001	(0.003)	−0.006	(0.008)	−0.018	(0.007)
Bankruptcy (−3; +3)	−0.023	(0.004)	−0.021	(0.004)	−0.015	(0.011)	−0.006	(0.009)
Foreclosure	0.036	(0.003)	−0.005	(0.003)	0.082	(0.005)	−0.002	(0.005)

Table 3: Price discounts and value components. Interactions reveal whether discounts are tied to building value, land value, or local market value.

TABLE 4—VAR FOR NEIGHBORHOOD HOUSE PRICES

	sf_t		Δpf_t		$pu_t - pf_t$		Δpu_t	
	Estimate	Std err	Estimate	Std err	Estimate	Std err	Estimate	Std err
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A. VAR</i>								
Δpf_{t-1}	—	—	−0.072	(0.012)	0.038	(0.012)	−0.033	(0.004)
$pu_{t-1} - pf_{t-1}$	—	—	0.840	(0.016)	0.067	(0.016)	−0.093	(0.006)
$\Delta \text{Adj } R^2$			0.384		0.144		0.016	
<i>Panel B. VAR with lagged forced share</i>								
sf_{t-1}	0.596	(0.012)	−0.633	(0.070)	0.575	(0.070)	−0.057	(0.027)
Δpf_{t-1}	−0.004	(0.002)	−0.067	(0.012)	0.034	(0.012)	−0.033	(0.004)
$pu_{t-1} - pf_{t-1}$	0.006	(0.003)	0.857	(0.016)	0.051	(0.016)	−0.092	(0.006)
$\Delta \text{Adj } R^2$	0.368		0.390		0.152		0.016	
<i>Panel C. VAR with lagged forced share interactions</i>								
sf_{t-1}	0.548	(0.016)	0.166	(0.097)	0.320	(0.098)	0.486	(0.037)
Δpf_{t-1}	0.011	(0.003)	−0.044	(0.015)	0.041	(0.015)	−0.003	(0.006)
$pu_{t-1} - pf_{t-1}$	0.007	(0.003)	0.984	(0.020)	0.018	(0.020)	0.002	(0.008)
$sf_{t-1} \times \Delta pf_{t-1}$	−0.263	(0.026)	−0.082	(0.153)	−0.202	(0.154)	−0.284	(0.057)
$sf_{t-1} \times (pu_{t-1} - pf_{t-1})$	0.038	(0.031)	−1.989	(0.181)	0.565	(0.181)	−1.424	(0.068)
$\Delta \text{Adj } R^2$	0.373		0.400		0.153		0.054	

Notes: Table reports estimates and standard errors, in parentheses, from vector auto-regressions (VAR) of percentage change in average forced and unforced house prices at the ZIP code-year level. The average price of forced sales is represented by pf , pu is the average price of unforced sales, and sf is the share of forced sales in each ZIP code at time t . Each specification includes neighborhood and time fixed effects. The number of observations in each regression is 7,254. The reported “ $\Delta \text{Adj } R^2$ ” is the difference between the adjusted R^2 of the full model and the adjusted R^2 of the model with only time dummies.

Table 4: VAR for neighborhood house prices. The VAR shows mean reversion in forced-sale prices and limited predictability of unforced-sale prices.

- The same variables add little explanatory power for unforced-price growth; the added adjusted R^2 for unforced growth is only 0.016 in panels A and B.
- Panel B shows that the forced-sale share is persistent (0.596) and depresses forced-price growth (−0.633), but has little effect on unforced-price growth.
- Panel C shows that high forced-sale shares make forced-sale discounts more persistent.

Interpretation: Table 4 addresses whether forced sales reveal future market prices. The answer is mostly no at the ZIP level. When the forced-sale discount is unusually large, it tends to close through a rebound in forced-sale prices rather than a fall in unforced-sale prices. This supports the view that unforced transactions are closer to efficient market prices, while forced-sale prices reflect time-varying illiquidity.

Economic message: Forced-sale prices are not simply better price discovery. They often look like bid-side prices in an illiquid market, and the forced-sale discount behaves like a temporary spread.

6.5 Table 5: Spillover Estimates of Foreclosures

Read:

- Table 5 estimates the effect of nearby foreclosures on transaction prices within a 0.25-mile radius.
- The close foreclosure slope is much stronger than the far slope: in column 1, close is −0.087 while far is −0.017.
- Columns 3 and 4 compare foreclosures before the sale to foreclosures after the sale. The close before-after difference is −0.020 in column 3 and −0.017 in column 4.
- Outlier controls show concavity: the marginal effect of additional foreclosures declines in the extreme right tail.
- The estimates are strongest at very local distances, consistent with neighborhood damage or highly local price externalities rather than broad ZIP-level price discovery.

Interpretation: Table 5 is the main spillover evidence. The paper does not have a perfect foreclosure instrument, so it uses timing and geography. Future nearby foreclosures proxy for unobserved local distress.

If past close foreclosures matter more than future close foreclosures, and close foreclosures matter more than far foreclosures, the pattern is more consistent with a causal local spillover.

Economic message: Foreclosure externalities are highly localized. A foreclosure wave may not move an entire ZIP code uniformly, but it can damage prices for very nearby houses.

TABLE 5—SPILLOVER ESTIMATES OF FORECLOSURES

	Using only foreclosures before transaction: Before $[\delta_{F,B}]$ and $\delta_{C,B}]$		Estimated difference in coefficients: Before–after $[(\delta_{F,B} - \delta_{F,A})$ and $(\delta_{C,B} - \delta_{C,A})]$	
	(1)	(2)	(3)	(4)
Slope: far (δ_F)	−0.017 (0.001)	−0.011 (0.001)	−0.006 (0.005)	−0.003 (0.001)
Slope: close (δ_C)	−0.087 (0.003)	−0.072 (0.003)	−0.020 (0.001)	−0.017 (0.003)
Outlier controls:				
Slope at 99.0: far ($\delta_F^{99.0}$)	0.002 (0.002)	−0.000 (0.002)	−0.011 (0.004)	−0.007 (0.003)
Slope at 99.0: close ($\delta_C^{99.0}$)	−0.055 (0.012)	−0.050 (0.011)	−0.048 (0.017)	−0.043 (0.014)
Slope at 99.5: far ($\delta_F^{99.5}$)	−0.004 (0.002)	−0.003 (0.002)	−0.008 (0.009)	−0.005 (0.002)
Slope at 99.5: close ($\delta_C^{99.5}$)	−0.037 (0.007)	−0.030 (0.006)	−0.031 (0.003)	−0.027 (0.008)
Slope at 99.9: far ($\delta_F^{99.9}$)	−0.001 (0.002)	−0.001 (0.002)	−0.001 (0.003)	−0.001 (0.002)
Slope at 99.9: close ($\delta_C^{99.9}$)	−0.009 (0.003)	−0.005 (0.003)	0.001 (0.002)	0.002 (0.004)
Additional controls:				
Average price, before		0.248 (0.002)		0.180 (0.002)
Average price, after				0.184 (0.002)
No transaction before indicator		2.992 (0.028)		2.168 (0.022)
No transaction after indicator				2.244 (0.022)

Notes: Table reports estimates and standard errors, in parentheses, from regressions of log price on the unweighted number of foreclosures in the 0.25-mile area around the house sold (variable *far*), and the linearly weighted number of foreclosures in the 0.1-mile area (variable *close*), for the year before and after the sale. The effect is specified as piecewise linear in the intervals (0–99th pct), (99th–99.5th), (99.5th–99.9th), (99.9th–max), with the estimated coefficients reported. Columns 1 and 2 are models that use only foreclosures that happened before each sale. The reported estimates are the slope coefficients of each part of the piecewise linear function. Columns 3 and 4 include the foreclosures before and after the sale. The reported estimates are the difference in the estimates for each piece of the piecewise linear function, between the effect of foreclosures before and after the transaction. Columns 2 and 4 also include the distance-weighted average log price of neighboring houses (0.25mi), in the

Table 5: Spillover estimates of foreclosures. The table compares effects of nearby foreclosures before and after the sale, with separate close and far measures.

7 Short summary

- The paper studies forced house sales in Massachusetts from 1987 through 2009 Q1.
- It identifies forced sales linked to deaths, bankruptcies, and foreclosures using administrative event data.
- Forced sales account for 6.1 percent of transactions, with foreclosures becoming dominant in 2008–2009.
- Hedonic regressions with census tract-year fixed effects show large forced-sale discounts.
- Foreclosures have the largest discount, roughly 27 percent in price terms.
- Death-related sales have smaller discounts, about 5–7 percent, and bankruptcy-related sales about 3 percent.
- Mechanism tests suggest death discounts are partly maintenance-related, bankruptcy discounts are liquidity-related, and foreclosure discounts reflect urgency plus vacancy/vandalism.

- ZIP-level VARs show forced-sale prices mean-revert, while unforced-sale prices are much harder to predict.
- This implies forced-sale discounts usually close through forced-sale price recovery, not by unforced prices falling.
- Very local foreclosure spillovers are negative and strongest within 0.1 mile.
- The spillover evidence is based on timing and distance comparisons, not a perfect foreclosure instrument.
- The main policy implication is that foreclosures can impose localized externalities on neighbors even when aggregate neighborhood price dynamics look stable.

8 Conclusion

8.1 Core conclusion

Campbell, Giglio, and Pathak show that forced sales in housing markets occur at sizable discounts. The magnitude of the discount depends strongly on the reason for the sale. Foreclosures are much more severely discounted than death- or bankruptcy-related sales, suggesting that lender liquidation, vacancy, and property deterioration are central to foreclosure pricing.

8.2 Economic intuition

Housing is not a liquid financial asset. A vacant house produces no housing services, requires maintenance, is costly to protect, and must be matched with a buyer who values its unique features. When a seller is forced to sell quickly, the seller may accept a low bid rather than wait for a better match. This creates a housing-market analog of a bid-ask spread.

8.3 Interpretation of the discount estimates

The forced-sale discount is not purely a measure of bargaining urgency. Death-related discounts appear partly tied to poor maintenance, especially for older sellers. Bankruptcy-related discounts look more like financial pressure. Foreclosure discounts reflect a combination of urgency, lender ownership, vacancy, possible vandalism, and local market distress.

8.4 Identification takeaway

The paper is strongest where it combines multiple pieces of evidence. The hedonic design controls for local market conditions, Table 3 tests mechanisms through value components, the VAR distinguishes forced-price mean reversion from unforced-price predictability, and Table 5 uses timing and distance to assess spillovers. No single design is fully experimental, but together they provide a coherent picture of forced-sale illiquidity and foreclosure externalities.

8.5 Policy implication

Foreclosure prevention and neighborhood stabilization policies may have benefits beyond the foreclosed property itself. If foreclosures lower nearby values, they can worsen neighbor equity positions and potentially create foreclosure feedback loops. The paper's local spillover evidence provides a microeconomic rationale for such concerns.

8.6 Limitations

The main limitation is that foreclosure timing and location are endogenous to neighborhood economic distress. The authors do not have a clean instrument for foreclosure incidence. Their causal interpretation relies on comparing past and future foreclosures and on the much stronger effects of very close foreclosures. The results are therefore best understood as strong suggestive evidence, not as a randomized experiment.

8.7 Takeaway

The paper's lasting message is that housing illiquidity is economically important. Forced sellers receive lower prices, foreclosures are especially costly, and foreclosure waves can create localized price externalities. The real estate market does not behave like a frictionless asset market when sellers are urgent and assets are fragile.